Technical Report on `4.ipynb`: Strict Fixed-Baseline-Template Sorting on Compressed-Reconstructed Data

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1 Objective

This report explains what `4.ipynb` does and analyzes its results. The notebook evaluates how well spike sorting can be performed on compressed/reconstructed data when using *fixed templates* learned from baseline (uncompressed) data. A key design choice is that baseline spike times are **not** used for candidate generation during strict sorting; baseline data is used only for evaluation alignment.

2 Data and Inputs

The notebook uses:

- Baseline Kilosort outputs from `F:/academic/.test_data/kilosort4`: templates, ops, spike times, and spike clusters.
- Reconstructed compressed binary data (float32), primarily `whitened_recon_ratio_0.10.bin` with fallback to `0.20` if missing.
- Channel overlap handling: baseline templates may differ from reconstructed channel count; the code uses shared channels (typically 383).

3 Methodology in the Notebook

3.1 Baseline template loading and preprocessing

The notebook loads baseline templates of shape `(267, 61, 383)` and baseline spike labels. The compressed/reconstructed inputs are derived from the upstream preprocessed signal saved in `whitened_data.npy`, where the signal has already undergone **CAR, bandpass filtering, and spatial whitening** in that order. Reconstructed data is then memory-mapped and reshaped to samples $\times$ channels (reported as `(1,350,000, 383)`).

3.2 Fixed-template matching setup

Kilosort template-matching tensors are prepared:

- Waveform PCA basis `wPCA` from baseline `ops`.
- Projected template tensor U via Einstein summation.
- Matching structure `ctc` from `ks_tm.prepare_matching`.

3.3 Strict full-timeline sorting (no baseline anchoring)

The notebook processes the entire reconstructed recording in windows (`win=12000`) with overlap (`2*nt`) to stabilize edge behavior. For each window:

- Run `ks_tm.run_matching` on channel-overlap data.
- Convert local detections to absolute times.
- Keep detections only in center region to avoid duplicates across overlaps.

Then concatenate and globally sort all detections by time.

Observed output in the notebook:

- Strict sorted spikes: **123,348**
- Unique predicted templates: **265**

3.4 Evaluation against baseline

The notebook evaluates strict-sort detections against baseline using nearest-neighbor time matching within ± 12 samples.

Reported for the strict run (ratio 0.10 section):

- Matched to baseline within ± 12: **121,700 / 123,348** (**98.66%**)
- Label agreement on matched spikes: **12.55%**

Interpretation: temporal alignment to nearest baseline events is high, but cluster-ID agreement is low under strict fixed-template transfer.

3.5 Cluster-level summary export

The notebook computes per-baseline-cluster:

- `n_ref`
- `n_matched_from_test`
- `same_label_rate`
- `recall_from_test_side`

and exports to `week4_fixed_template_cluster_summary.csv`.

This reveals substantial heterogeneity across clusters: some clusters have strong temporal recall but weak label consistency, suggesting template confusion or unstable identity transfer after compression/reconstruction.

3.6 Multi-ratio comparison (0.10, 0.20, 0.30, 0.50, 0.70)

If a ratio-specific reconstructed bin is missing, the notebook rebuilds it by:

1. DCT per channel on the upstream CAR-corrected, bandpass-filtered, and whitened signal,
2. keeping first $n\_\text{keep} = \lfloor n\_\text{samples} \cdot \text{ratio} \rfloor$ coefficients,
3. IDCT reconstruction,
4. writing `whitened_recon_ratio_X.bin`.

The notebook includes Windows file-lock-safe replacement logic (releasing active memmaps and retrying `os.replace`) to avoid `PermissionError: WinError 32` during overwrite.

For each ratio, events are evaluated at baseline times (same-event protocol), producing:

Ratio	n_{base}	n_{det}	Det. Rate	Cls. Acc. (detected)	Overall Same-Event Acc.
0.10	137378	15848	0.1154	0.2383	0.0275
0.20	137378	15570	0.1133	0.2357	0.0267
0.30	137378	15583	0.1134	0.2333	0.0265
0.50	137378	15896	0.1157	0.2331	0.0270
0.70	137378	15938	0.1160	0.2339	0.0271

Table 1: Multi-ratio fixed-template same-event evaluation (from `week4_fixed_template_ratio_compare.csv`).

4 Result Analysis

4.1 Main findings

1. **Detection rate is low and stable across ratios.** Around 11.3%–11.6% baseline events are detected in same-event evaluation.
2. **Classification accuracy on detected events is modest.** Around 23.3%–23.8%.
3. **Overall same-event accuracy is very low.** Around 2.65%–2.75%.
4. **Compression ratio has weak impact in this range.** Performance differences among 0.10–0.70 are small relative to absolute error.

4.2 Why strict timeline matching and same-event scores differ

The strict full-timeline run reported high nearest-baseline temporal matching (98.66%), but same-event evaluation at baseline times gives low detection/label scores. These two evaluations answer different questions:

- **Strict timeline nearest-neighbor match**: "Can detected spikes be temporally mapped to some nearby baseline event?"
- **Same-event baseline-time protocol**: "At each baseline event, does reconstructed-data matching detect and classify the same event correctly?"

The second criterion is stricter and exposes identity-transfer weakness.

4.3 Implications

The current fixed-template transfer appears robust for producing many plausible detections in time, but weak for preserving exact cluster identity under reconstruction perturbations. This suggests:

- template mismatch after compression/reconstruction,
- insufficient discriminability with fixed baseline templates,
- possible sensitivity to thresholding / overlap resolution / channel harmonization.

5 Limitations and Next Steps

1. Refit or adapt templates on reconstructed data (domain adaptation) rather than pure fixed-template transfer.

2. Add confusion-matrix analysis between baseline and predicted template IDs to quantify systematic remapping.
3. Tune tolerance, matching thresholds, and windowing for sensitivity/specificity trade-offs.
4. Evaluate precision-recall style metrics under one-to-one event assignment constraints.
5. Test whether channel-standardization strategy (383 vs broader mappings) materially changes identity accuracy.

6 Produced Files by the Notebook

- `week4_kilosort_function_fixed_template_results.npy`
- `week4_fixed_template_cluster_summary.csv`
- `week4_fixed_template_ratio_compare.csv`
- ratio-specific reconstructed bins `whitened_recon_ratio_*.bin`